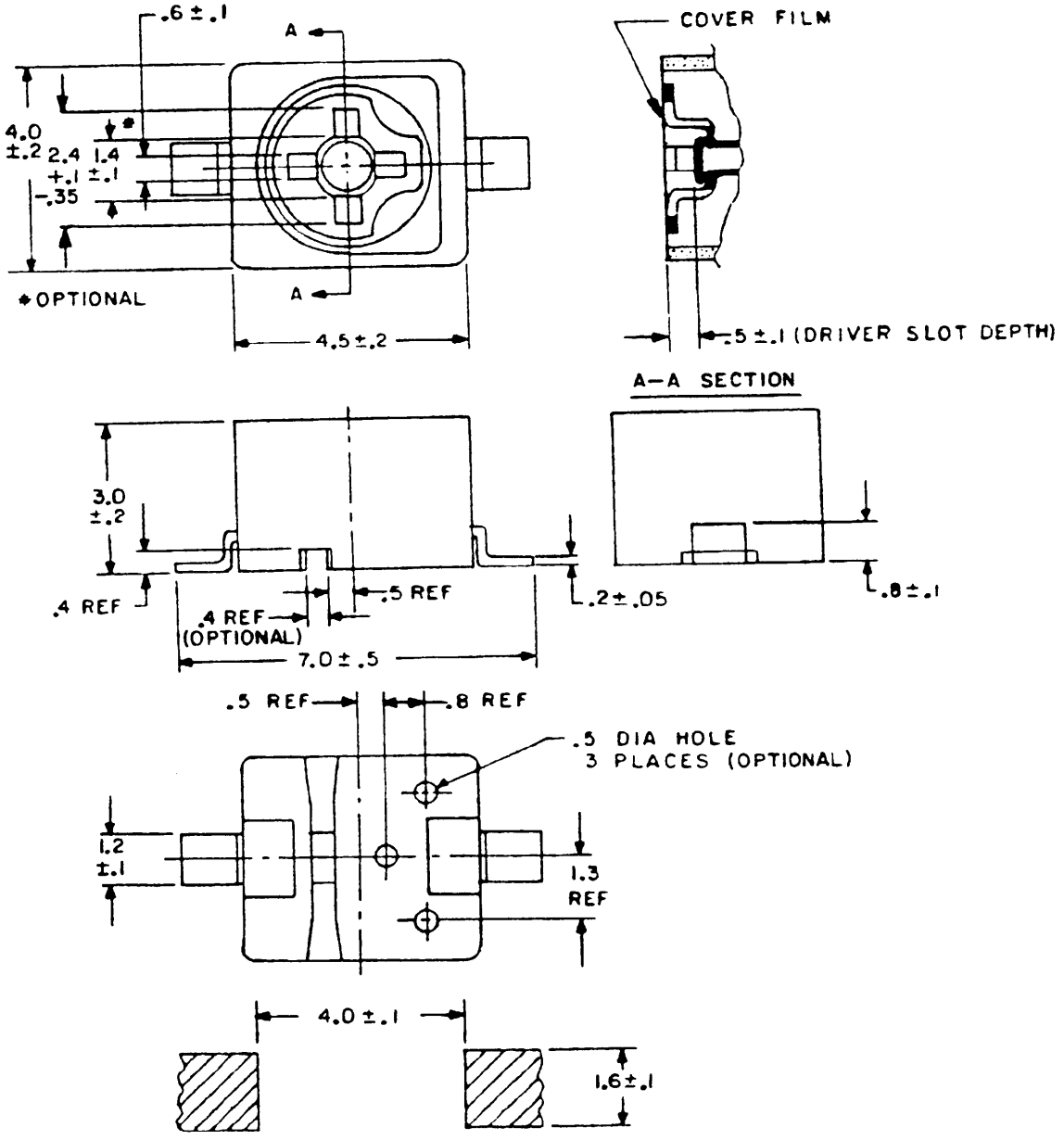
MILITARY SPECIFICATION SHEET

CAPACITORS, VARIABLE, CERAMIC DIELECTRIC, STYLES CV50 AND CV51

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-81.



RECOMMENDED PATTERN ON P.C.B.

Style CV50

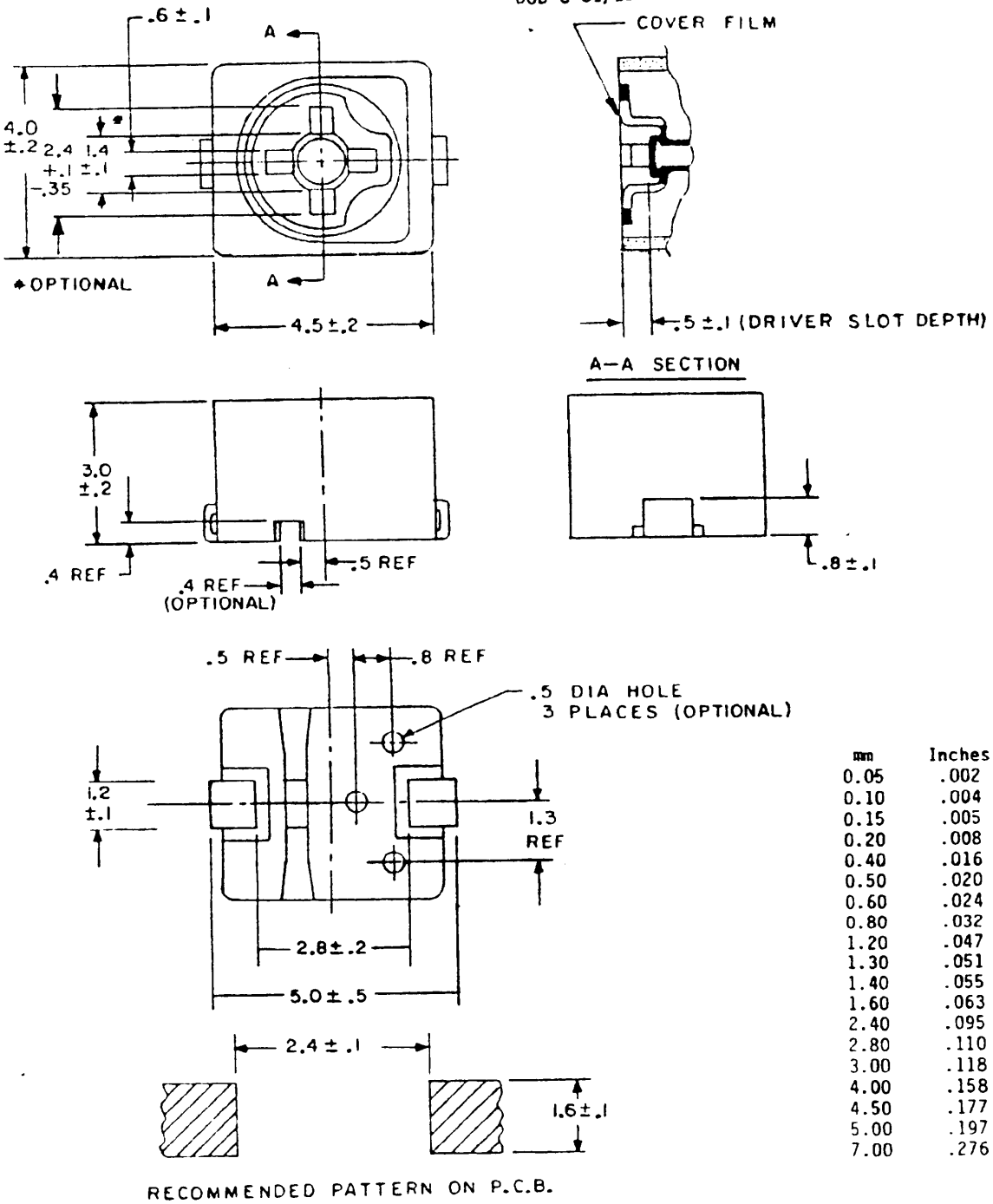
FIGURE 1. Dimensions and configuration.

FSC 5910

AMSC N/A

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RECOMMENDED PATTERN ON P.C.B.

mm	Inches
0.05	.002
0.10	.004
0.15	.005
0.20	.008
0.40	.016
0.50	.020
0.60	.024
0.80	.032
1.20	.047
1.30	.051
1.40	.055
1.60	.063
2.40	.095
2.80	.110
3.00	.118
4.00	.158
4.50	.177
5.00	.197
7.00	.276

NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.15.
4. This unit is supplied with a silicone membrane covering the adjustment opening which allows washing prior to tuning without risk of contamination.

Style CV51

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Capacitance value: See table I.

DC rated voltage: 100 volts dc.

Operating temperature range: -55°C to +85°C

Characteristic: See table I.

Dielectric withstanding voltage: Method 301 of MIL-STD-202. A dc potential of 2.2 times rated voltage applied between terminals for 3 ±2 seconds.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition D (100,000 ft.).

Test potential: 100 percent of dc rated voltage.

Insulation resistance: Method 302 of MIL-STD-202, condition A. 100 volts dc applied - 10,000 megohms, minimum.

Capacitance: Method 305 of MIL-STD-202.

Dissipation factor: At 1 MHz ±100 KHz, at maximum and minimum capacitance and shall not be more than 0.2 percent for rated maximum capacitance 6 pF, 10 pF, and 20 pF, 0.3 percent for 30 pF.

TABLE I. Styles CV50 and CV51

Type designation 1/ 2/	Capacitance		DC rated voltage	Characteristic	
	Min	Max		Symbol	Temperature coefficient
	pF	pF	volts		ppm/°C
CV5XX060	2.0	6.0	100	X	0 ±200
CV5XT100	3.0	10.0	100	T	-150 ±300
CV5XZ200	4.5	20.0	100	Z	-750 ±300
CV5XV300	6.5	30.0	100	V	-1200 ±500

1/ The complete type designation will include an additional digit to indicate style CV 50 or CV 51.

2/ Qualification to the CV50 will also qualify CV51.

Temperature coefficent: Within the limits specified for the applicable characteristic (see table I).

Capacitance drift: Within 0.75 percent of initial step 1 measurement or 0.50 picofarad (pF), whichever is greater.

Terminal strength:

Pull test: Capacitor held by body and 4 pound load applied to each terminal for at least 10 seconds.

NOTE: This test is applied for leaded type only.

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Torque: Not less than .2 nor more than 1.4 ounce inches.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100 G).

Vibration, high frequency: Method 204 of MIL-STD-202, condition B (15 G).

Capacitance change: Shall not exceed ±2 percent of nominal value or 0.5 pF, whichever is greater.

Dissipation factor: Shall be not more than 0.2 percent for rated maximum capacitance 6 pF, 10 pF, and 20 pF, 0.3 percent for 30 pF.

Dielectric withstanding voltage: A dc potential of 2.2 times rated voltage applied between terminals for 3 ±2 seconds.

Insulation resistance: 10,000 megohms, minimum.

Fatigue: At a rate of approximately 20 cycles per minute for a period of 1 minute.

ΔC: Shall not exceed 12 percent or 0.75 pF, whichever is greater.

Torque: Initial requirement.

Life:

Qualification test: 1,000 hours at +85°C, 150 percent of rated volts dc with a peak alternating voltage of 50 percent of rated volts dc (100 hertz or less) superimposed.

Insulation resistance: Initial requirement.

Capacitance change: Shall not exceed ±8 percent of nominal value or 0.75 pF, whichever is greater.

Group C life: Conditions and requirements are the same as that for qualification.

Moisture resistance: Method 106 of MIL-STD-202.

Insulation resistance: 10,000 megohms minimum.

Capacitance change: Shall not exceed ±5 percent of nominal value or 0.5 pF, whichever is greater.

Dissipation factor: Shall be not more than 0.5 percent.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 11, 80, 99
DLA - ES

User activity:
Army - AR, AT
Air Force - 19

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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